

UNIT - I

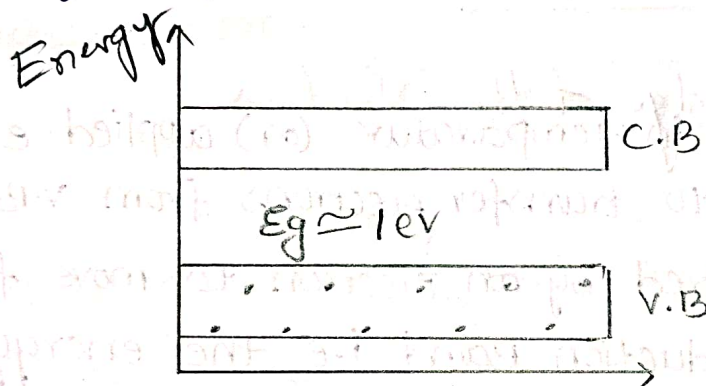
(1)

Explain the Semiconductor, Insulators and metals classify using energy band Diagram.

Based on the electrical conductivity all the materials in nature are classified as Semiconductors, Insulators and metals.

Semiconductors :-

- * Semiconductor is a material that has its conductivity somewhere between the insulator and conductor.
- * The forbidden gap between valency Band and Conduction Band typically ranges from 1eV.

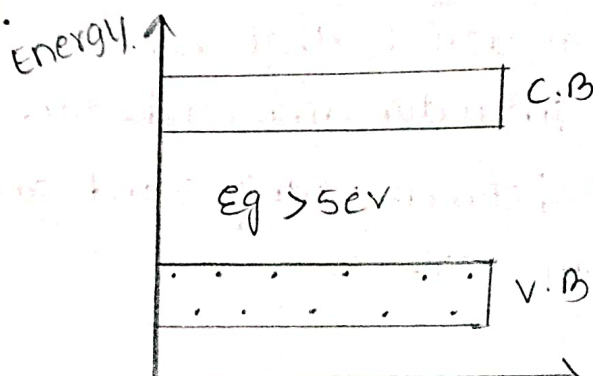


- * At 0K and at low temperatures, the valence Band electrons do not have sufficient energy to move from V.B to C.B.
- * Thus, Semiconductors act as insulators at 0K.
- * As the temperature increases, a large number of valence e^- acquire sufficient energy to leave the V.B, cross the forbidden bandgap and reach C.B.
- * These are now free electrons as they can move freely under the influence of electric field.

(2)

Insulators:- An insulator is a material that offers a very low level of conductivity when voltage is applied.

- * Typical resistivity level of an insulator is of the order of 10^{10} to $10^{12} \Omega\text{-cm}$.
- * There is a large forbidden gap between V.B and C.B.
- * The forbidden gap between V.B and C.B is greater than 5 eV.

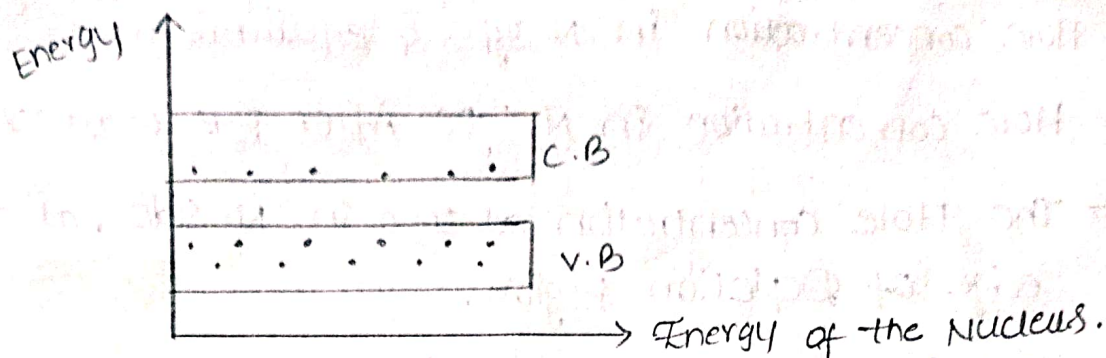


- * Even an increase in temperature (or) applied electric field is insufficient to transfer electrons from V.B to C.B.
 - * The energy required by an electron to move from valence band to conduction band i.e. the energy required for a valence electron to become a free electron.
 - * Electrons are free to accelerate under the influence of external voltage source resulting in the flow of charge.
- Ex:- wood, Glass, paper, quartz.

Metals:-

- * A metal is a material that offers a conductivity without applying any energy.
- * In metals, the V.B and C.B either overlap (or) the valence band itself is partially filled.

(3)



* There is no forbidden energy gap, electrons can easily transition to higher energy states and participate in conduction.

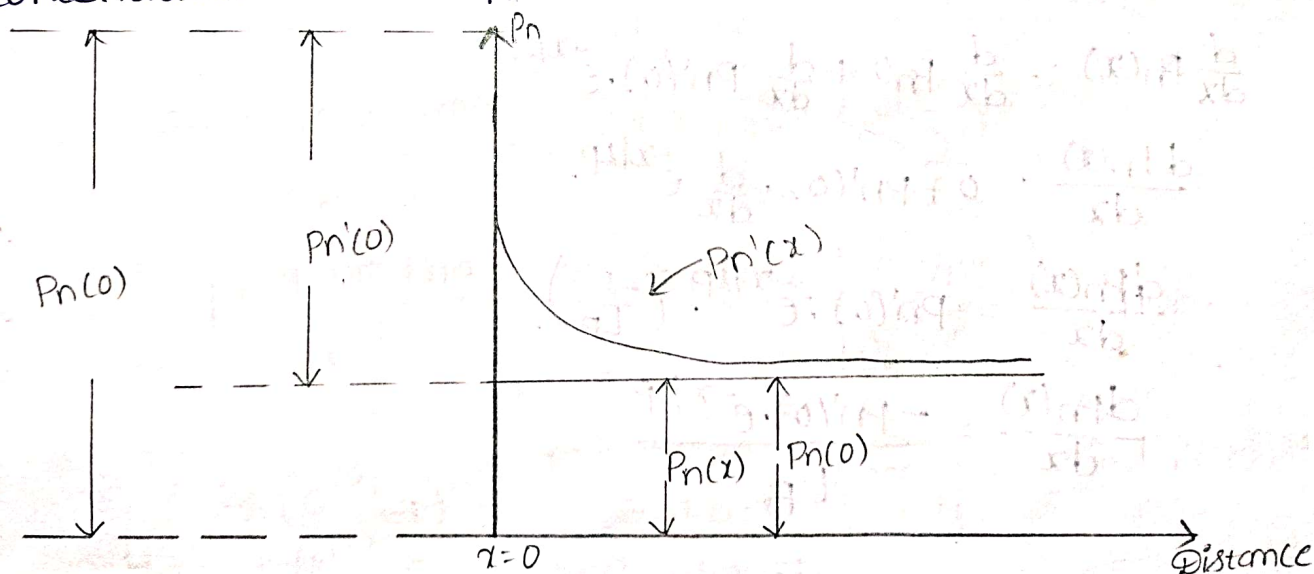
* The Fermi level lies within the conduction band or the overlapping region of the valence and conduction bands.

* That free electrons are always available for conduction, even at absolute zero.

Ex:- Al, Cu, Fe, Ag

2. Derive the Expression for the diode current equation.

* The Law of Junction gives the variation of carrier concentration w.r.t applied voltage.



(4)

P_{n0} = Hole concentration in N-type & equilibrium.

$P_n(x)$ = Hole concentration in N-type After F-B w.r.t 'x'.

$P_n(0)$ = The Hole concentration at $x=0$ in N-side, at the edge of Depletion Region.

$P_n'(x)$ = Injected hole concentration in N-side w.r.t 'x'.

$P_n'(0)$ = Injected hole concentration in N-side at $x=0$.

The Injected hole concentration in N-side w.r.t 'x' is,

$$P_n'(x) = P_n(x) - P_{n0} = P_n'(0) \cdot e^{-x/L_p} \quad \text{--- (1)}$$

$$P_n(x) = P_{n0} + P_n'(0) \cdot e^{-x/L_p} \quad \text{--- (2)}$$

at $x=0$

$$P_n'(0) = P_n(0) - P_{n0} \quad \text{--- (3)}$$

The diffusion current of holes in N-side is given as,

$$I_{Pn}(x) = -Aq \cdot D_p \frac{dP_n(x)}{dx} \quad \text{--- (4)}$$

A_q = Area, D_p = Diffusion co-efficient of holes.

Diff eqn (2) w.r.t 'x'

$$\frac{d}{dx} P_n(x) = \frac{d}{dx} P_{n0} + \frac{d}{dx} P_n'(0) \cdot e^{-x/L_p}$$

$$\frac{dP_n(x)}{dx} = 0 + P_n'(0) \cdot \frac{d}{dx} e^{-x/L_p}$$

$$\frac{dP_n(x)}{dx} = P_n'(0) \cdot e^{-x/L_p} \cdot \left(-\frac{1}{L_p}\right)$$

$$\frac{dP_n(x)}{dx} = \frac{-P_n'(0) \cdot e^{-x/L_p}}{L_p}$$

sub above eqn in eqn (4)

$$I_{Pn}(x) = -Aq \cdot D_p \left[\frac{-Pn'(0) \cdot e^{-x/L_p}}{L_p} \right]$$

$$I_{Pn}(x) = \frac{Aq \cdot D_p \cdot Pn'(0) \cdot e^{-x/L_p}}{L_p}$$

at $x=0$,

$$I_{Pn}(0) = \frac{Aq \cdot D_p \cdot Pn'(0) \cdot e^{-x/L_p}}{L_p} \quad \text{--- (5)}$$

Similarly, the diffusion current of $\bar{e}$ in p-side is given as

$$I_{np}(0) = \frac{Aq \cdot D_n \cdot np'(0) \cdot e^{-x/L_n}}{L_n} \quad \text{--- (6)}$$

w.k.t, $I = I_{Pn}(0) + I_{np}(0)$.

$$I = \frac{Aq \cdot D_p \cdot Pn'(0) \cdot e^{-x/L_p}}{L_p} + \frac{Aq \cdot D_n \cdot np'(0) \cdot e^{-x/L_n}}{L_n}$$

from Law of Junction,

$$Pn'(0) = Pn0 \cdot (e^{V/V_T} - 1)$$

$$np'(0) = np0 (e^{V/V_T} - 1)$$

$$I = \frac{Aq \cdot D_p \cdot Pn0 \cdot e^{V/V_T}}{L_p} + \frac{Aq \cdot D_n \cdot np0 (e^{V/V_T})}{L_n}$$

$$I = \left[\frac{Aq \cdot D_p \cdot Pn0}{L_p} + \frac{Aq \cdot D_n \cdot np0}{L_n} \right] (e^{V/V_T} - 1)$$

I_0

$$I = I_0 (e^{V/V_T} - 1) \Rightarrow \boxed{I = I_0 \cdot e^{V/u \cdot V_T} - 1}$$

$u=1$, for Ge | $u=2$, for Si.

(6)

3. Draw and explain the VI characteristics of PN junction diode in both forward and reverse Bias.

Forward Bias:-

* When a diode is connected in a Forward Bias Condition, a negative voltage is applied to the N-type material and a positive voltage is applied to the p-type material.

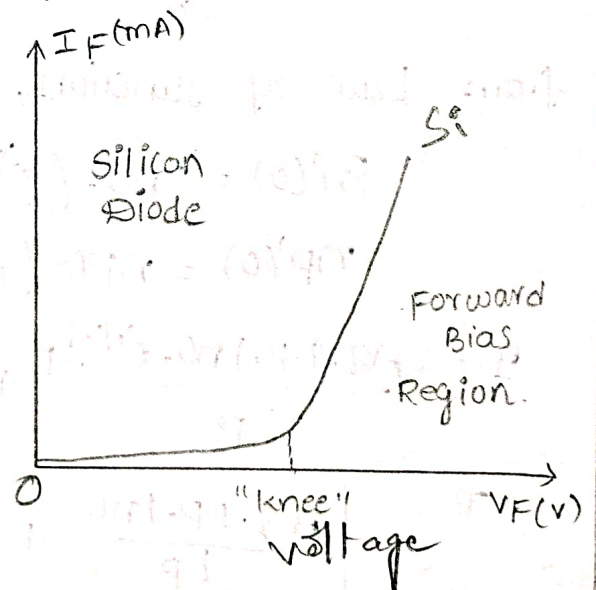
* If this external voltage becomes greater than the value of the potential, approx. 0.7 volts for silicon and 0.3 volts for germanium, the potential barriers opposition will be overcome and current will start to flow.

* The characteristics curve of zero current flowing up to this voltage point, called the "Knee" on the static curves and then a high current flow through the diode with little increase in the external voltage.

* Current remains negligible until the threshold voltage ($\sim 0.7V$ for silicon).

* After threshold, the current increases exponentially.

* The point at which this sudden increase in current takes place is represented on the static $V-I$ characteristic curve above as the "knee" point.



Cut in voltage
(or)
Threshold voltage

Reverse
Bias
max

Reverse Bias :- when a diode is connected in a Reverse Bias Condition, a positive voltage is applied to the N-type material and a negative voltage is applied to the p-type material.

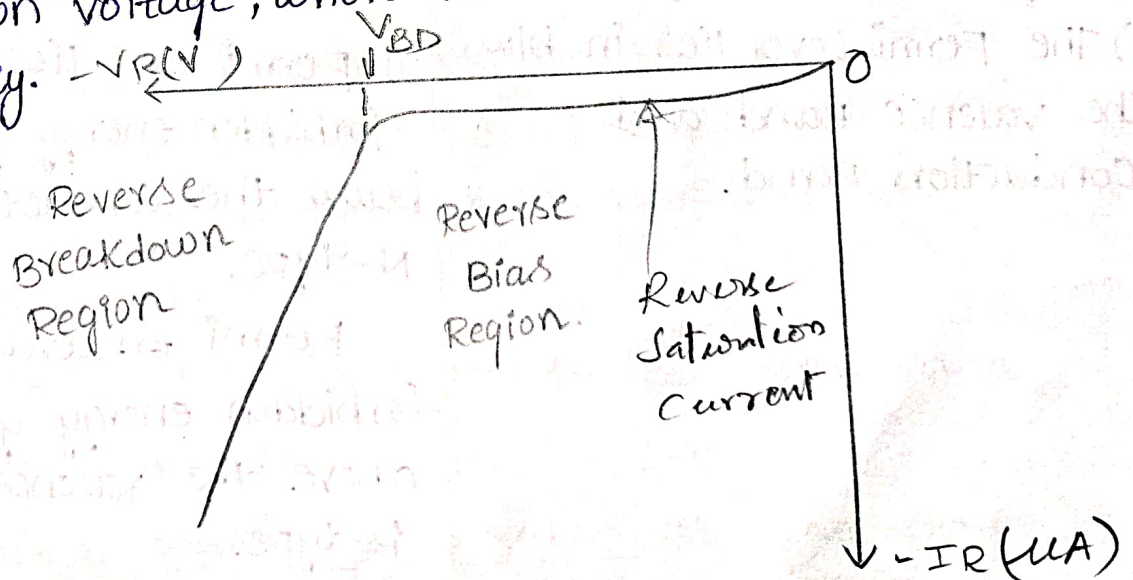
* The positive voltage applied to the N-type material attracts electrons towards the positive electrode and away from the junction, while the holes in the p-type end are also attracted away from the junction towards the negative electrode.

* A high potential barrier is created thus preventing current from flowing through the semiconductor material.

* If the reverse voltage exceeds a critical value called the breakdown voltage, the current increases sharply.

* At the breakdown voltage, the current increases significantly due to avalanche (or) Zener Breakdown.

* Stays almost flat (leakage current) until breakdown voltage, where current increases rapidly.



4. Compare Intrinsic and Extrinsic Semiconductors.

Intrinsic Semiconductor	Extrinsic Semiconductor.
1) A Semiconductor in its pure form without impurity is called an Intrinsic semiconductor. 2) The number of free electrons is equal to the number of holes. 3) The conductivity is low. 4) In these, the Fermi energy level lies in the middle of valence band and conduction band. 5) Its electrical conductivity dependence on temperature only. 6) Exhibits both n-type and p-type conductivity. 7) The Fermi Level lies in b/w the valence band and conduction band.	1) when a small amount of impurity is doped in pure form of semiconductor it becomes extrinsic semiconductor. 2) The number of free electrons and hole never equal. 3) The conductivity is high. 4) In these, the Fermi energy level shifts towards valence or conduction energy band. 5) Its electrical conductivity depends on temperature as well as on the amount of impurity. 6) Exhibits n-type or p-type conductivity based on doping. 7) Fermi level lies in the forbidden energy gap slightly below the conduction band, in N-type.
	Fermi level lies in the forbidden energy gap slightly above the valence band in p-type.

Intrinsic Semiconductor.

- 8) In these, the Resistivity is High.
- 9) In these, No additional Donor Level.
- 10) Not commonly used for practical devices.
- 11) Ex: Silicon (Si), Germanium (Ge)

Extrinsic Semiconductor.

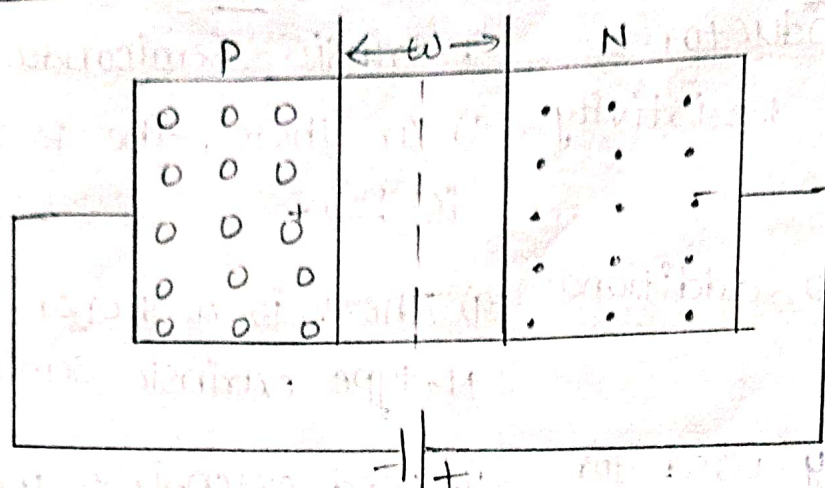
- 8) In these, the Resistivity is Low.
- 9) There is a Donor Level in N-type extrinsic semiconductor.
- 10) used extensively in electronic devices like diodes (or) transistors.
- 11) Ex: N-type :- Phosphorus (P), Antimony (Sb)
P-type :- Boron (B), Indium (In)

5. Explain the terms Transition capacitance and Diffusion Capacitance.

Transition Capacitance :-

- 1) The transition capacitance occurs at Reverse Biase Condition.
- 2) when p-N junction is reverse biased the depletion region acts as an insulator (or) as a Dielectric medium and the p-type and n-type region have low resistance and act as the plates.
- 3) Thus this p-N Junction can be considered as a parallel plate capacitor.
- 4) This Junction capacitance is called as space charge Capacitance (or) Transition capacitance and is denoted as C_T .

(10)



5) Reverse Bias causes the majority charge carriers to move away from the junction, so the thickness of the depletion region denoted as " w " increases with the increase in reverse bias voltage.

6) The formula for transition capacitance is given as C_T ,

$$C_T = \frac{A \cdot \epsilon}{w}$$

where, A = cross sectional area of the region.
 w = width of the depletion region.

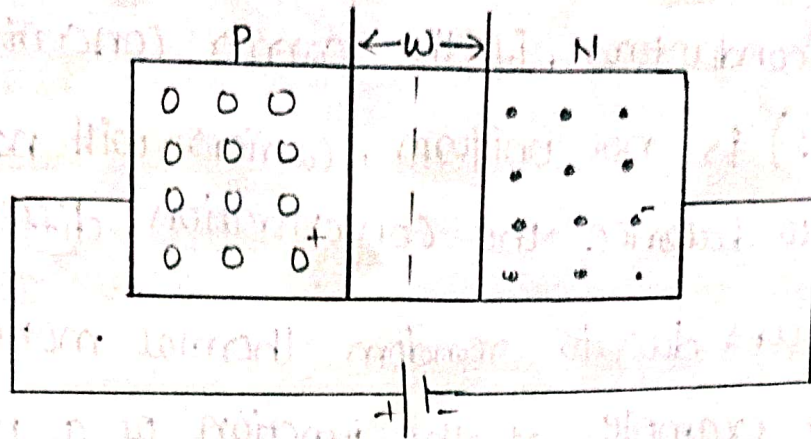
7) when the depletion region increases the resulting transition capacitance decreases.

Diffusion capacitance :-

1) The Diffusion Capacitance occurs at Forward Bias Condition.

2) when the junction is forward biased, a capacitance comes into play, that is known as Diffusion capacitance denoted as C_D . It is much greater than the Transition Capacitance.

3) The Density of the charge carriers is high near the junction and reduces as the distance increases.



4) charge is stored on both side of the junction and varies with the applied potential. so as per definition change in charge with respect to applied voltage results in capacitance which here is called as Diffusion Capacitance.

5) The formula for Diffusion capacitance is $C_D = \frac{\tau \cdot I}{\eta \cdot V_T}$

τ - mean life time of charge carriers

I - Diode current, V_T - applied forward voltage.

η - Generation recombination factor.

6) The Diffusion capacitance is directly proportional to the Diode current.

7) In forward biased $C_D \gg C_T$. And thus C_T can be neglected.

6) Explain the Diffusion and drift currents for a Semiconductor.

Diffusion current :- Diffusion is the movement of particles from a region of high concentration to a region of low concentration.

* In semiconductors, if the carrier concentration (electrons or holes) is not uniform, carriers will naturally spread out to balance the concentration difference.

* This occurs due to random thermal motion of the carriers. For example, at the junction of a p-type and n-type semiconductor, electrons on the n-side will diffuse to the p-side.

* The Diffusion current Density ($J_{\text{diffusion}}$) is given by:

$$J_{\text{diffusion}} = qD \frac{dN}{dx}$$

Ex: In a p-n junction, electrons diffuse from the n-region to the p-region, while holes diffuse from the p-region to n-region, creating a Diffusion current.

Drift current :-

* Drift current is the motion of charge carriers due to an external electric field. The electric field applies a force on the carriers, causing them to move in a specific direction.

* When an electric field is applied across a Semiconductor, electrons move toward the positive terminal, and holes move toward the negative terminal, creating a current.

* The Drift current density (J_{drift}) is given by:

$$J_{\text{drift}} = qn\mu E$$

* In a reverse-biased p-n junction, the electric field sweeps electrons and holes toward opposite directions, causing a drift current.

Explain current components in PN Junction Diode?

* When a PN Diode is forward biased, the depletion region is very small and large forward current flows. The holes from p side are injected to n-side and electrons from n-side are injected to p side.

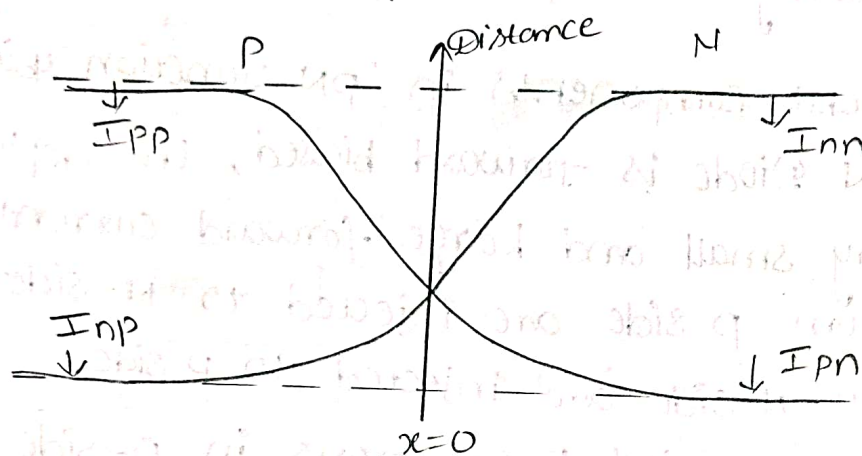
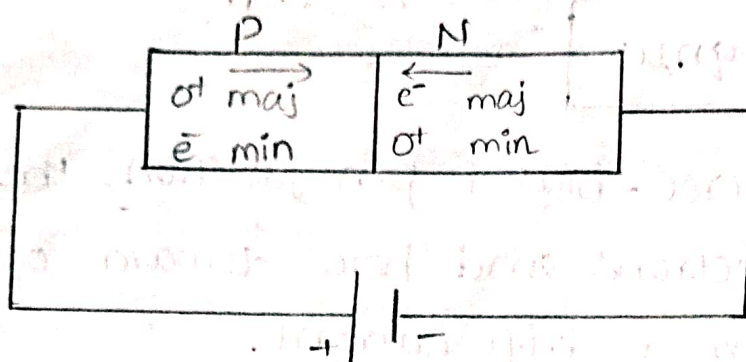
* Thus current carried by electrons in p-side, I_{np} is due to minority carriers and it decreases exponentially with respect to distance measured from the junction.

* Similarly, the current due to holes diffusing from p-side to n-side is due to minority carriers and decreases exponentially with respect to distance measured from junction.

* This current is represented as I_{pn} .

* At the junction $x=0$, the electrons constituting current in p-side is $I_{np}(0)$ and the holes crossing the junction from p-side to n-side is $I_{pn}(0)$. Thus, the current at the junction is the total current (I) flowing through the circuit and is given as.

$$I = I_{pn}(0) + I_{np}(0).$$



* on p-side the current component is due to holes on p-side which are majority charge carriers. It is denoted by $I_{pp}(x)$. The total current in p-side is written as

$$I = I_{pp}(x) + I_{np}(x).$$

* $I_{np}(x)$ = current due to electrons in p-side.

Similarly, the electron on n-side, which are majority carriers, constitute is the sum of current components due to e^- in n-side and holes in n-side.

$$I = I_{nn}(x) + I_{pn}(x)$$

where, $I_{pp}(x)$ - current due to holes in p-side.

$I_{nn}(x)$ - current due to electrons in n-side.

$I_{pn}(x)$ - current due to holes in n-side.

$I_{np}(x)$ - current due to electrons in p-side.